

RNA Biology and Muscle Development in Cerebral Palsy

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Financial Disclosures

Speaker Name: Mona Batish, Ph.D.

1. Disclosure of Relevant Financial Relationships

I have no financial relationships to disclose.

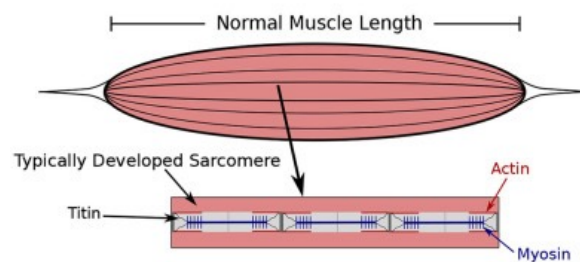
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I will not discuss off label use and/or investigational use in my presentation.

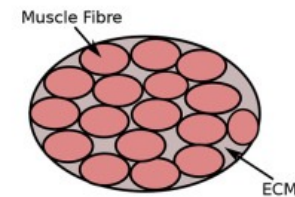
Muscle Development



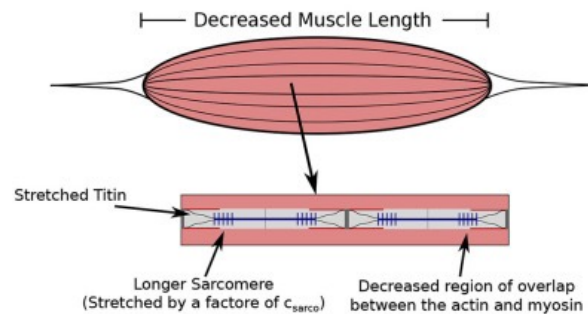
Typically Developed Muscle



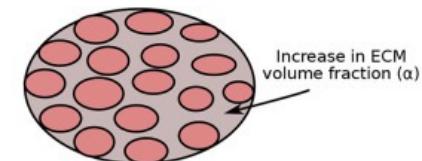
Muscle Cross-section



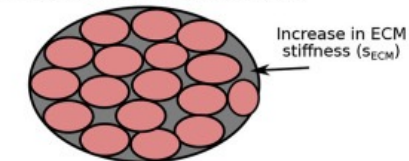
Cerebral Palsy Muscle



Fibrotic Muscle (CP)



Increased ECM Stiffness (CP)



Biomarker discovery in CP

Neuroimaging of CP

- Cranial ultrasound and brain MRI

Genetic biomarkers of CP

- A potential genetic etiology can be identified in 10-30% of individuals with CP

Epigenetic and Transcriptional biomarkers of CP

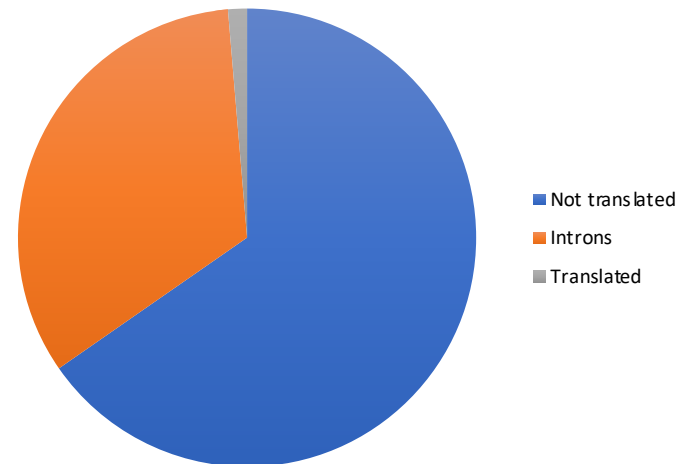
- DNA methylation
- Altered gene expression
- Non-coding RNAs

Protein as Biomarkers of CP

- Dysregulated proteome in CP
- Altered post translational modification
- Protein mis localization

Dark matter of genome: Noncoding RNAs

- Only 2% of human genome get translated to proteins
- Most of the remaining 98% does get transcribed
- About 50-70% of this encompasses intronic regions
- A large fraction of untranslated (including intronic) regions lead to a vast number of noncoding RNAs
- Different types of noncoding RNAs have been reported and still being identified including tRNAs, rRNAs, snoRNAs, miRNAs, lncRNAs, **circular RNAs** etc.



RNA is not just the messenger but also an end product molecule that regulate critical biological processes.

Roles of noncoding RNAs in CP

● human CP tissue or biofluid

● CP animal model

● a process central to CP

○ not tested in CP

microRNA

miRNA · about 22 nucleotides

Silences messages — targets an mRNA for degradation or blocks its translation.

The best-studied class, with the most CP-model evidence of the three.

- **miR-1**
raised in a rodent CP model; acts on neuron apoptosis and neural stem-cell differentiation
- **miR-26a**
alleviates brain injury in CP rats; inhibits astrocyte activation; aids axon growth via PTEN
- **miR-135b**
alleviates brain injury in hypoxic–ischemic CP rats; reported downregulated in CP
- **miR-200a**
reduces neural apoptosis in CP rats
- **miR-128-3p**
targets Smurf2/YY1 to reduce neuronal apoptosis and improve neurobehaviors in CP rats
- **miR-9-5p**
falls after hypoxia–ischemia; restoring it limits brain damage via DDIT4 autophagy

- **miR-486**
~2-fold higher in circulating vesicles in CP; targets PAX7, lowers sarcomerogenesis genes
- **miR-206**
myogenic differentiation via Pax7 and HDAC4; also repairs the neuromuscular junction
- **miR-29c**
stimulates satellite-cell proliferation; represses atrophy and matrix genes
- **miR-200c**
overexpression inhibits skeletal muscle differentiation
- **miR-1**
regulates myogenesis; implicated in Duchenne muscular dystrophy

Long noncoding RNA

lncRNA · longer than 200 nucleotides

Long transcripts that are never translated; several act as sponges.

- **MIAT**
mitigates neuron apoptosis in hypoxic–ischemic neonatal CP via miR-211/GDNF
- **MALAT1**
competing endogenous RNA with miR-1; neuroprotective, inhibits neuron apoptosis
- **hsrw**
neurodevelopment at the neuromuscular junction, by targeting FUS

- **MALAT1**
also regulates satellite-cell proliferation and differentiation
- **lncR-125b**
sponges miR-125b, raising IGF2 to trigger satellite-cell differentiation
- **linc-MD1**
times muscle differentiation; reduced in Duchenne muscle cells

Circular RNA

circRNA · covalently closed loops

Made by back-splicing; works mainly by sponging microRNAs and RNA-binding proteins.

Most recent class of non coding RNAs

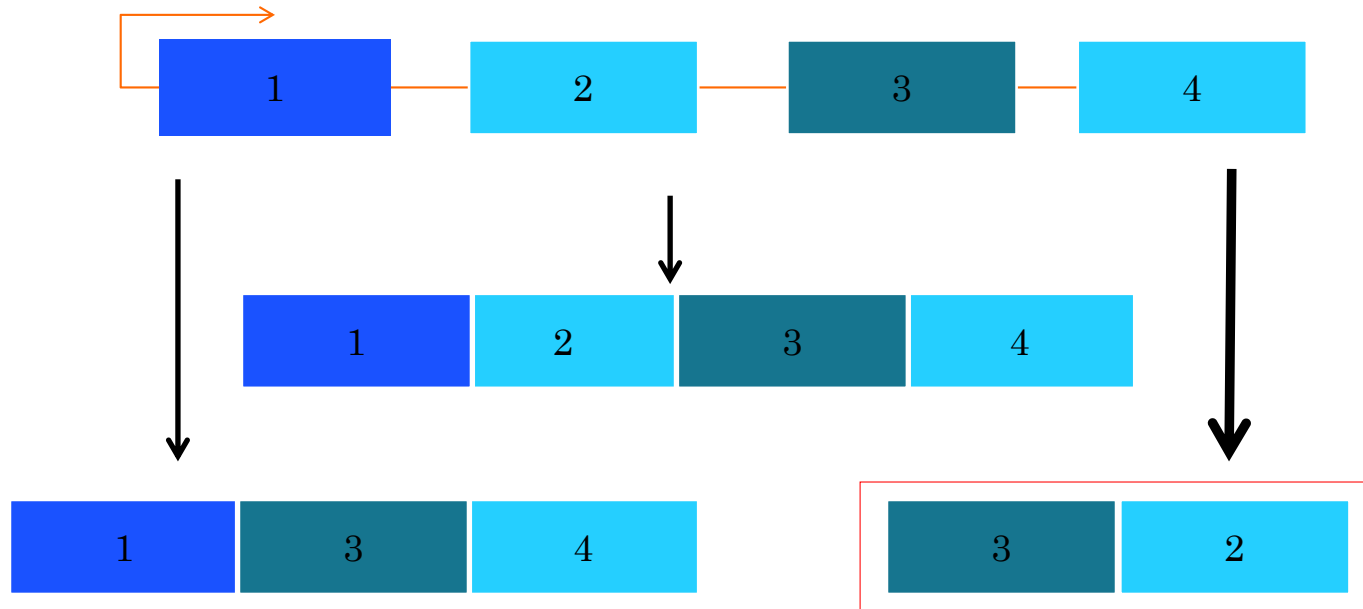
- **hsa_circ_0086354**
reduced in plasma from children with CP; AUC 0.967 in 30 case–control pairs at one center
- **Brain enrichment**
highly enriched in the brain and implicated in several neurological disorders
- **Epilepsy**
altered circRNA profiles in patients and in animal models
- **Biomarker potential**
proposed to predict cerebral palsy, its trajectory and treatment response

- **circNFIX**
reduced in CP satellite cells; sponges miR-373-3p to permit MEF2C and sarcomere genes
- **CDR1as**
raises IGF1R by sponging miR-7; MyoD-induced at the proliferation-to-differentiation switch
- **circSVIL, circFGFR2, circHIPK3, circRBF2s, circTMTCT1**
myogenesis — chicken
- **circLMO7, circFUT10, circFGFR4 · circZfp609**
myogenesis — bovine and mouse
- **circFAM188B**
encodes a peptide promoting proliferation while repressing myogenesis

IN THE DEVELOPING BRAIN

IN THE GROWING MUSCLE

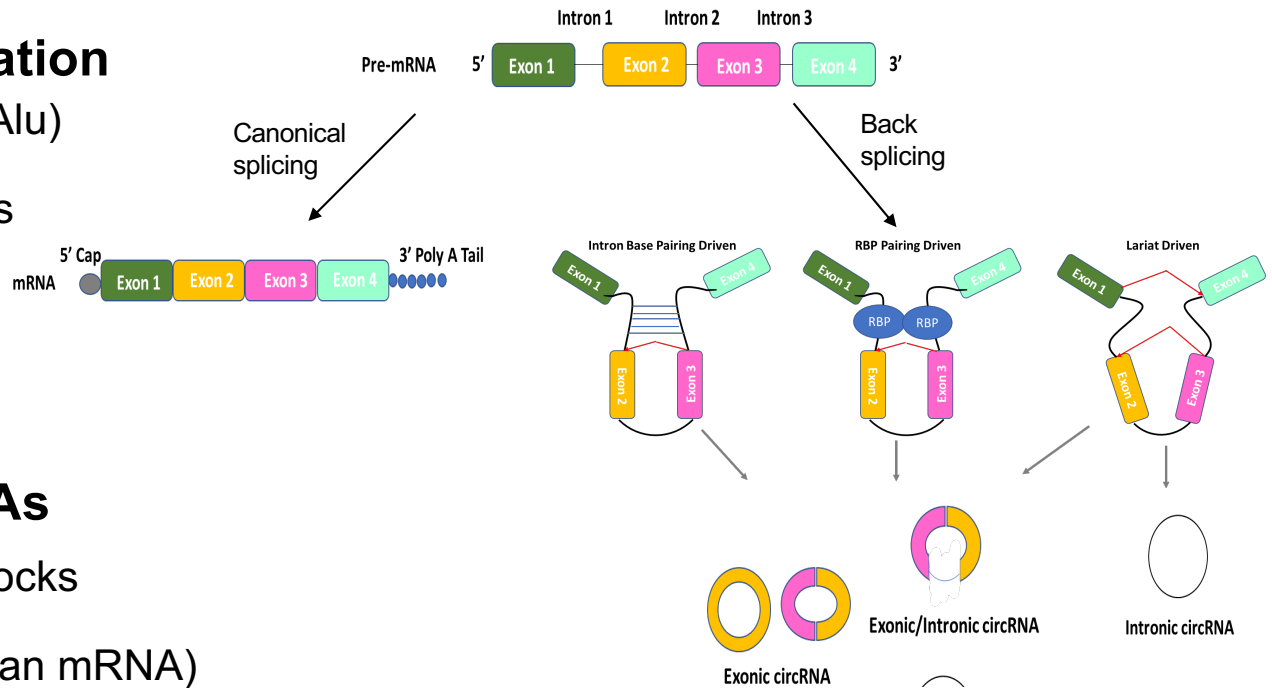
Circular RNAs: Mystery of scrambled exons



Biogenesis of circular RNAs

Features promoting circularization

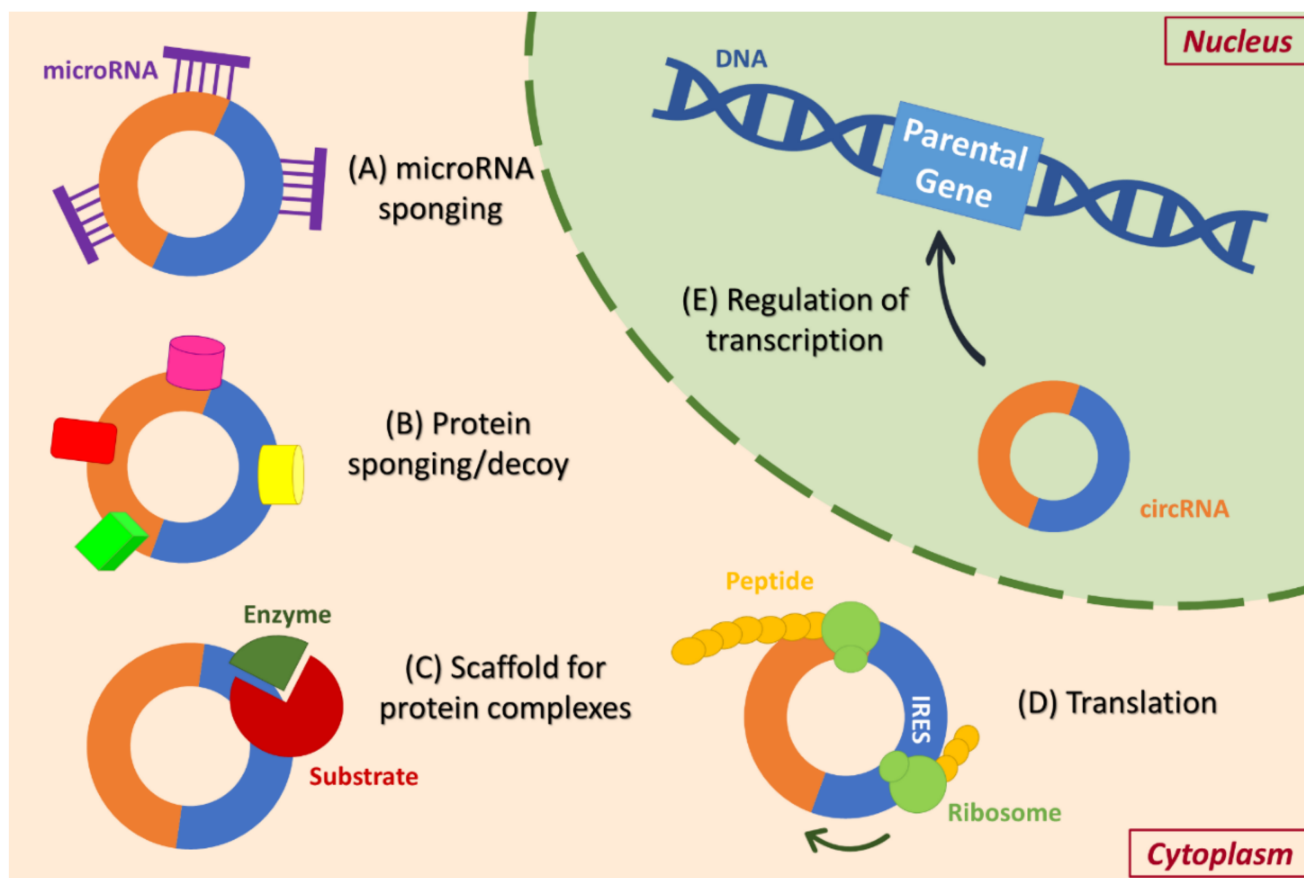
- Complementary intron sequences (Alu)
- Longer than average flanking introns
- Longer than average exons



Characteristics of circular RNAs

- Made from regular RNA building blocks
- Very stable (~5 time more stable than mRNA)
- ~1% from overall RNA
- Conserved across species
- Great candidates for non-invasive diagnosis

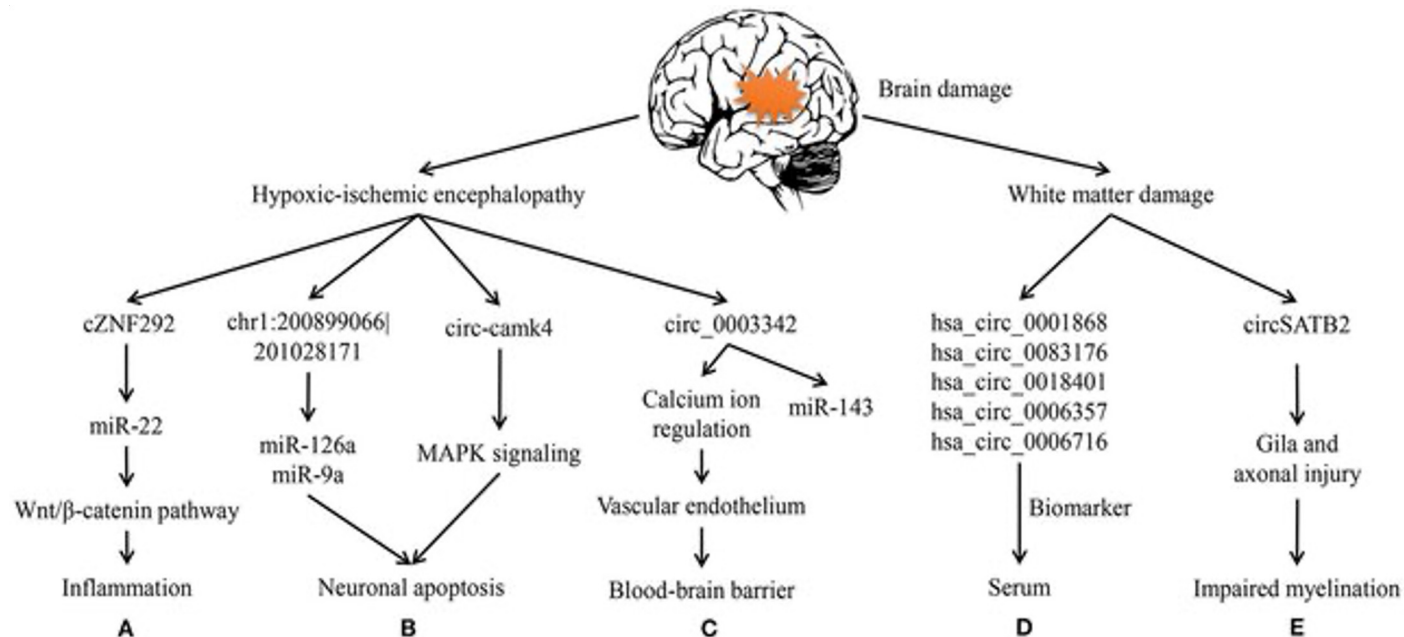
Functions of circular RNAs



Circular RNAs are most enriched in Brain and Muscle.

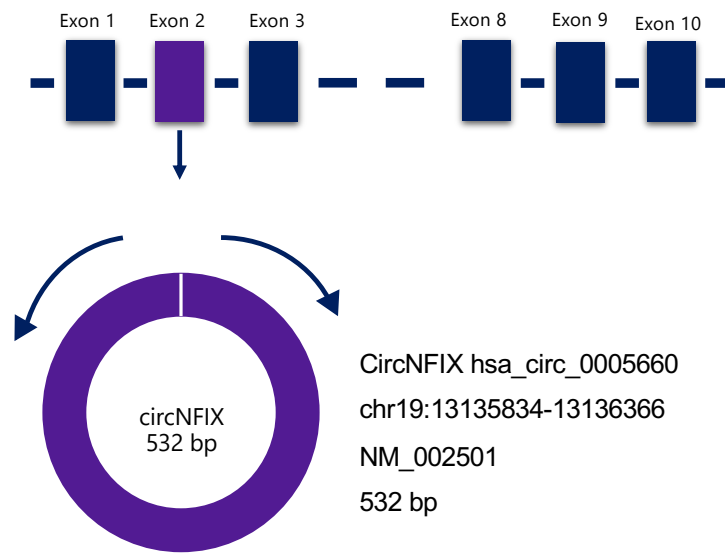
CircRNAs in brain injury and diseases

- Circular RNAs have a direct impact on neurodevelopment and brain injury.
- CircRNAs have potential applications as diagnostic biomarkers and as a target for neuroprotective intervention.



Role of circRNAs in Cerebral Palsy is just beginning to be appreciated!

Circular RNA Nuclear Factor IX - circNFIX



- CircNFIX participates in the development of many organs of the human body.
- **CircNFIX promotes MEF2C expression during muscle cell differentiation in mouse models**
- Myocyte-specific enhancer factor 2C (MEF2C) is a transcription factor that regulates skeletal muscle development, sarcomeric gene expression, fiber type control, and glucose metabolism.
- CircNFIX is evolutionarily conserved in mouse.
- It's abundant in the heart, and its overexpression inhibits cardiomyocyte proliferation.

circNFIX as a miRNA sponge in spastic CP

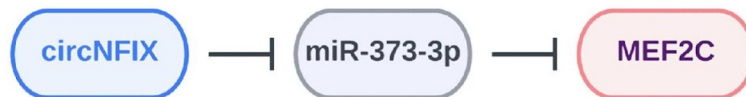
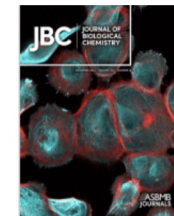
RESEARCH ARTICLE · Volume 300, Issue 12, 107987, December 2024 · Open Access

The circular RNA circNFIX regulates MEF2C expression in muscle satellite cells in spastic cerebral palsy

Brigette Romero¹ · Parsa Hoque¹ · Karyn G. Robinson² · ... · Vijay Parashar¹ · Robert E. Akins² · Mona Batish¹  

... Show more

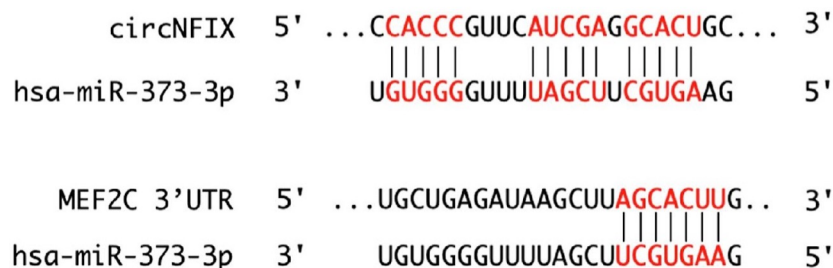
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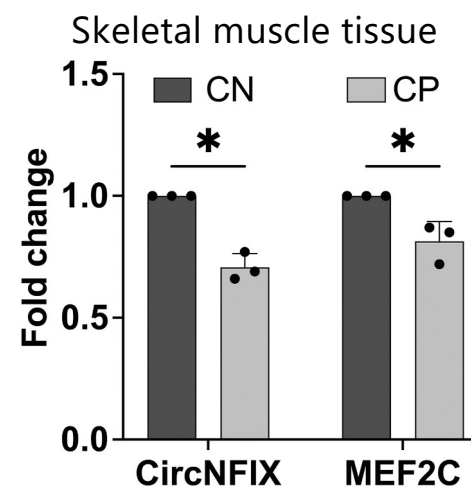
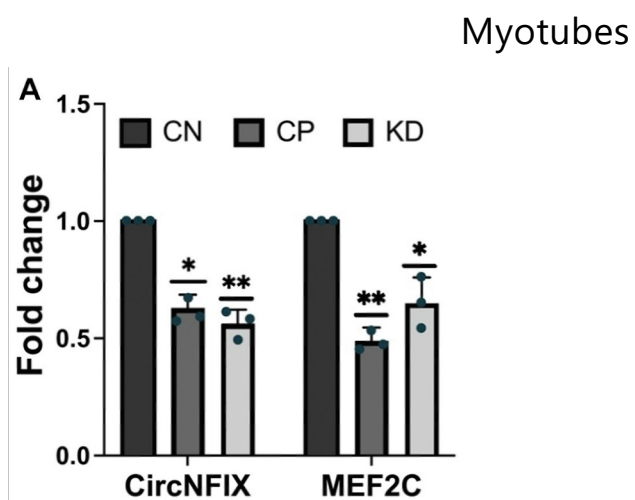
CircRNA:miRNA stoichiometry

CircNFIX: 64 copies/ cell

miR-373-3p: 43 copies/cell

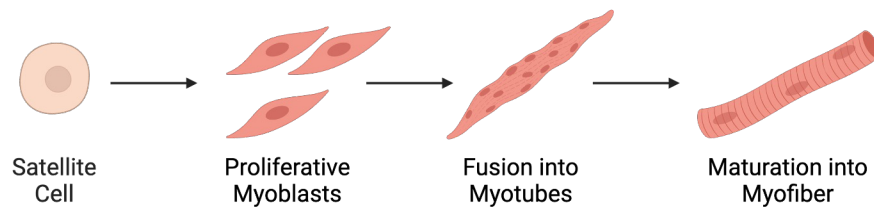


circNFIX regulates MEF2C expression in CP



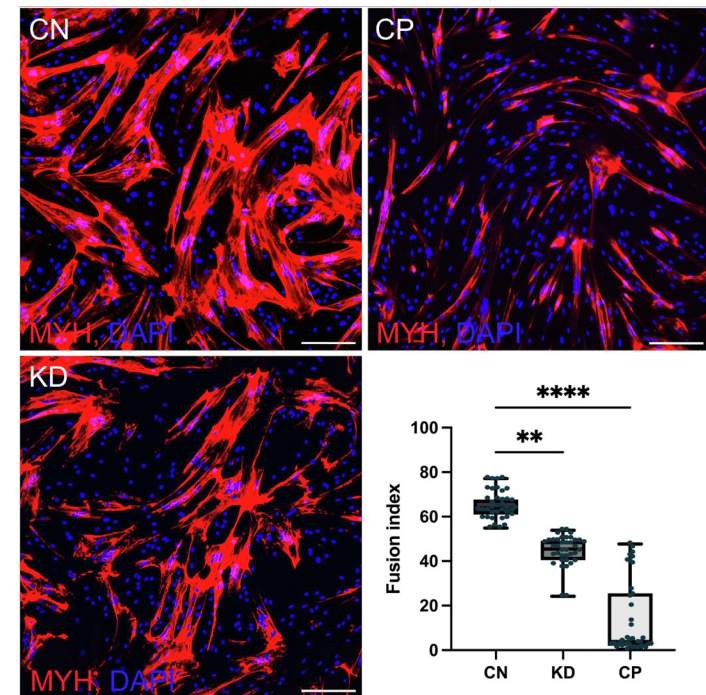
* = <0.05; ** = <0.01,
*** = <0.005

circNFIX loss leads to reduced fusion index



Created with BioRender.com

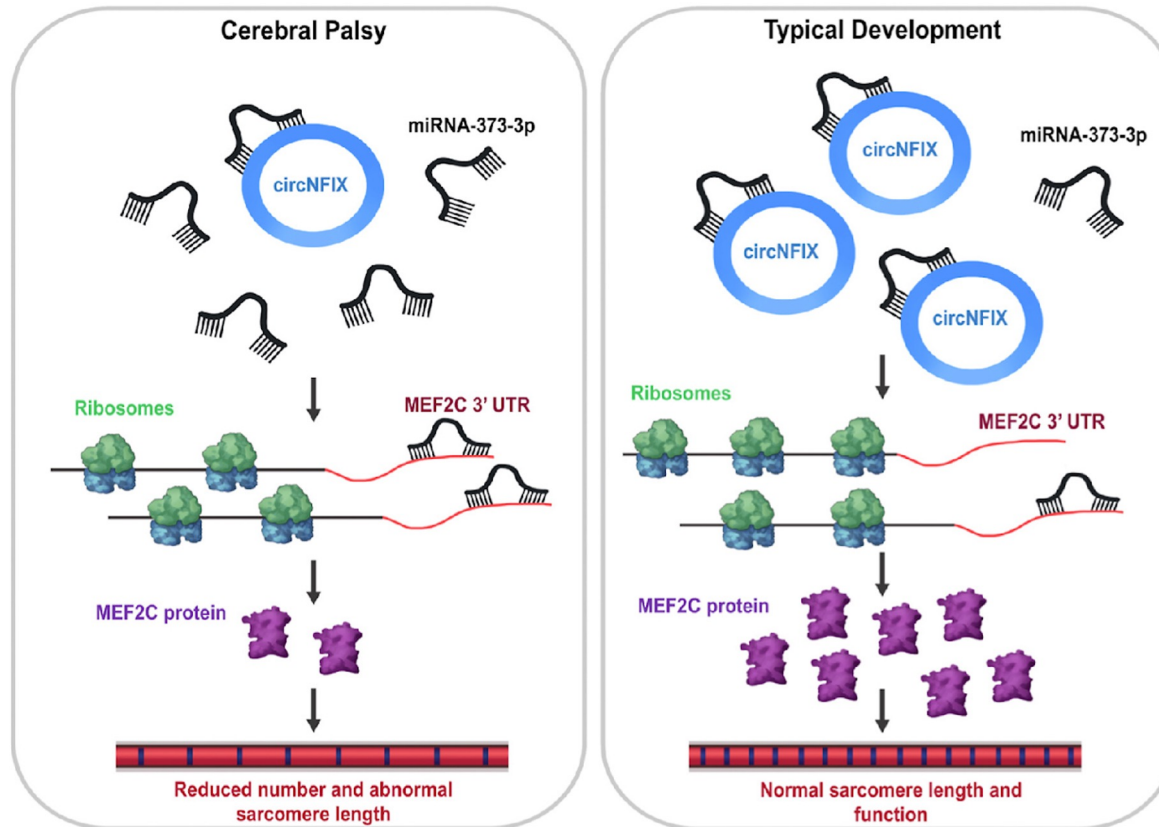
- The fusion index is the percentage of nuclei inside fused cells.
- CP myoblasts have a reduced ability to fuse and form mature muscle fibers, often resulting in weaker muscle function.



** = <0.05, **** = <0.0001

Romero, B et al., (2024). *JBC*, 300(12).

Model of the role of circNFIX in CP

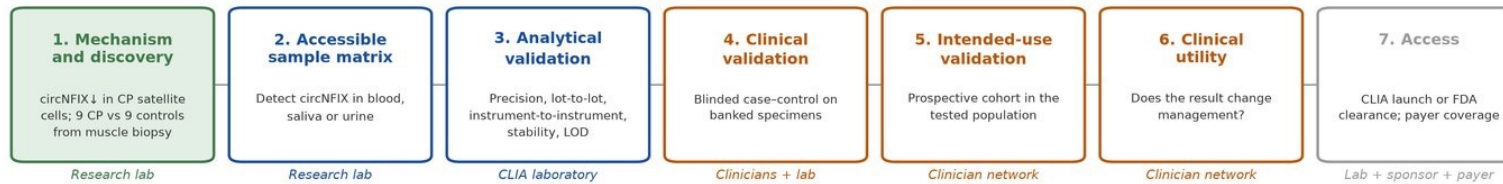


CircNFIX has a potential to be used as early biomarker for Cerebral Palsy!

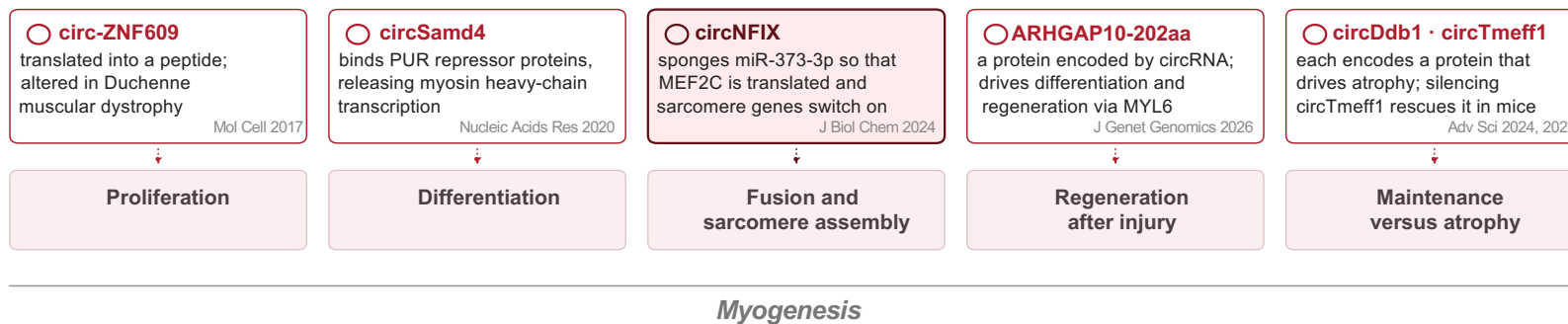
Romero, B et al., (2024). *JBC*, 300(12).

Can we use circRNAs as Biomarker?

circNFI



Circular RNAs that regulate act different steps of muscle formation



Circular RNAs that are reported to be altered in other human muscle diseases

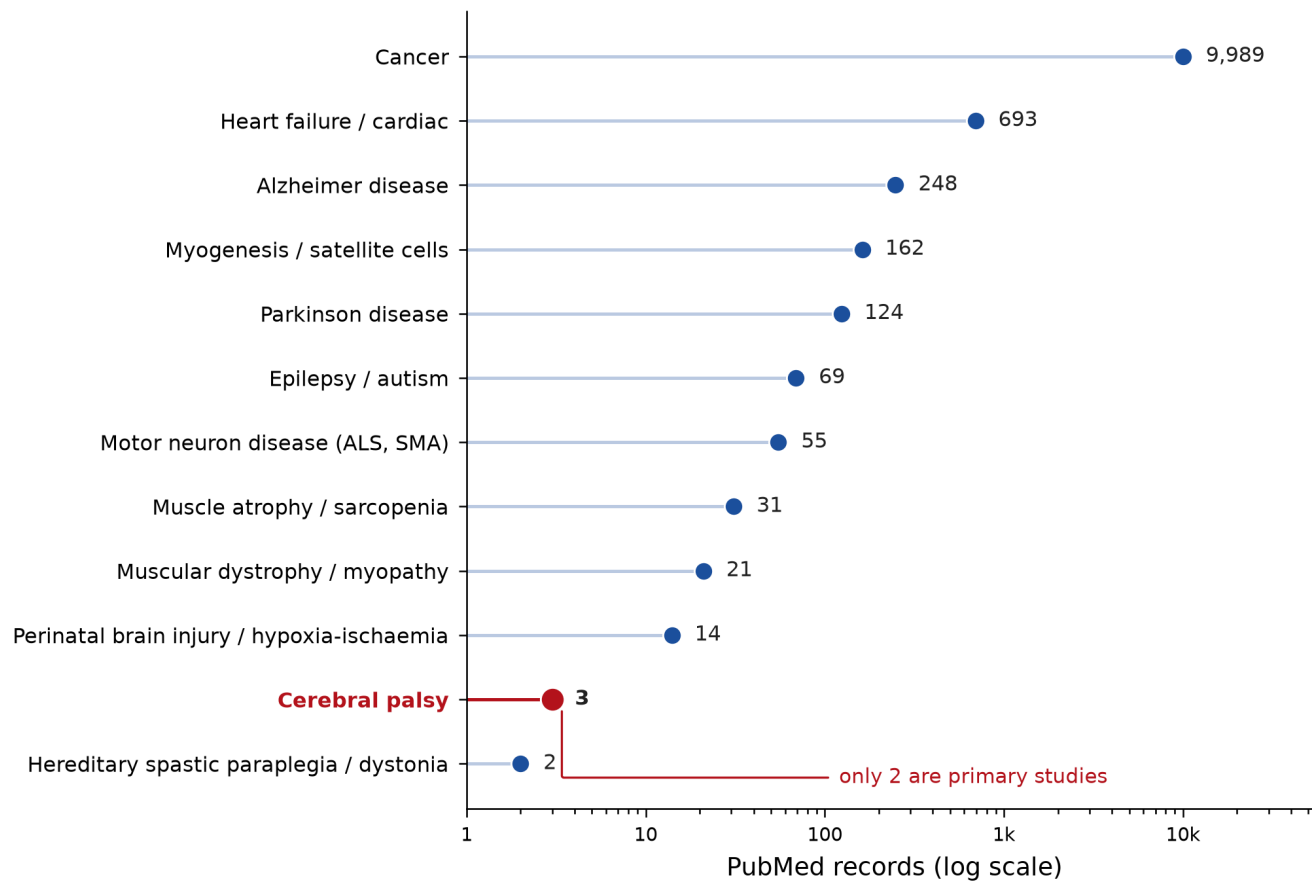
Duchenne muscular dystrophy (DMD)

197 circular RNAs are reported to be dysregulated in DMD

Myotonic dystrophy type 1

circARHGAP10 tracks CTG repeat length and runs inversely with muscle strength;

circRNAs in Diseases



Almost ten thousand circular RNAs have been reported in literature for cancer while only 3 for Cerebral Palsy

Can circRNAs be used for clinical tests?

Circulating circular RNAs have been shown to carry clinical-grade signal

Disease (sample type)	Description of the test	Reference
Alzheimer's Diagnosis assay (Blood)	34-circRNA panel; Discovery in 1221 individuals and the test is in preclinical stage	Phillips, B. et al. Blood-based circular RNAs for early diagnosis of Alzheimer's disease. <i>Nat. Med.</i> https://doi.org/10.1038/s41591-026-04485-5 (2026). PMID: 42387213.
Pancreatic cancer diagnosis (Plasma)	Circulating circRNA panel for early detection, built and tested across multiple international cohorts	Xu, C. et al. A Circulating Panel of circRNA Biomarkers for the Noninvasive and Early Detection of Pancreatic Ductal Adenocarcinoma. <i>Gastroenterology</i> . https://doi.org/10.1053/j.gastro.2023.09.050 (2024) PMID: 37839499
Myotonic Dystrophy Type1 (Muscle)	circARHGAP10 regulating the CTG repeat length and muscle strength	Baci, D. et al. circARHGAP10 as a candidate biomarker and therapeutic target in myotonic dystrophy type 1. <i>Mol Ther Nucleic Acids</i> . https://doi.org/10.1016/j.omtn.2025.102646 (2025). PMID: 40896579.

Circular RNAs can be accurately measured in clinical samples

Method type	Description of the method	Reference
CRISPR-Cas13 droplet ChIP (Biochemical)	Can detect 2 circRNAs from plasma of clinical samples using dual target single droplet quantification in 70 minutes	Phillips, B. et al. Blood-based circular RNAs for early diagnosis of Alzheimer's disease. <i>Nat. Med.</i> https://doi.org/10.1038/s41591-026-04485-5 (2026). PMID: 42387213.
CIRI3 (Bioinformatic)	Detection and quantification of circRNAs from RNA seq Datasets	Zheng, X. et al. Detecting and quantifying circular RNAs in terabyte-scale RNA-seq datasets with CIRI3. <i>Nat. Biotechnol.</i> https://doi.org/10.1038/s41587-025-02835-1 . (2026). PMID: 41034498.
Single-Molecule Imaging (Imaging)	Imaging, quantification and sub cellular localization	Koppula, A. et al. CircFISH: A Novel Method for the Simultaneous Imaging of Linear and Circular RNAs. <i>Cancers</i> . https://doi.org/10.3390/cancers14020428 . (2022). PMID: 35053590.

CircRNA is an ideal “theragnostic molecule”: The question is when will it reach Cerebral Palsy

Can we diagnose cerebral palsy earlier?

Why it matters?

- Median age of diagnosis is 18 months to 3 years
- Often flagged upon missing milestones
- Unmet need for specific and sensitive molecular biomarkers!

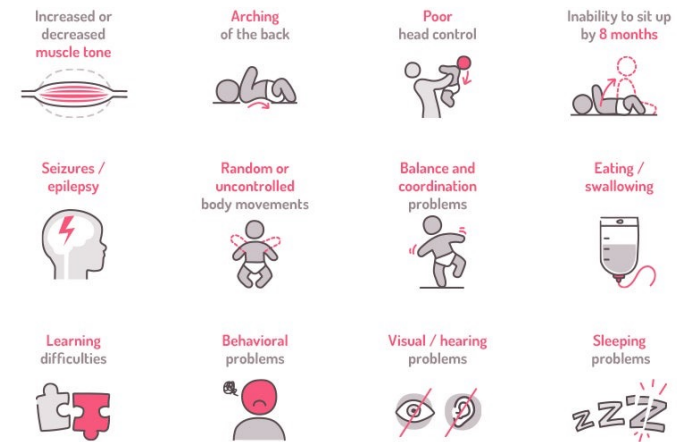
What a Molecular biomarker would have to do?

- Not replace those tools but extend existing diagnostics
- An objective affordable measure, in an accessible specimen (ideally noninvasive)
- Provide accurate diagnosis earlier than current timeline!

What will it take to get there?

- More research on molecular Biology behind CP
- Larger cohorts: longitudinal cohorts with banked, clinically annotated specimens
- Multi-omics: combining the knowledge of genomics, transcriptomics, epigenetics, and proteomics (in process)

Some early signs of Cerebral Palsy include:



<https://www.nestlehealthscience.com/mychildwithcerebralpalsy/hcp/cerebral-palsy>

Acknowledgments



Batish lab



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Robert Akins



Elizabeth Wright-Jin



Karyn Robinson



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- The Nemours CP Program
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- Swank Foundation
- American Academy for Cerebral Palsy and Developmental Medicine / Pedal with Pete Foundation
- The children and families served at Nemours



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Science
Foundation



Thank You !

A grayscale micrograph showing several large, roughly circular cells with thin outlines. The background is a fine, granular texture. In the bottom right corner, there is a horizontal scale bar labeled "1µm".

Thank You!

Comments?

Questions?

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